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Note: We are hoping to have our design be taped out

Mathematical Implementation

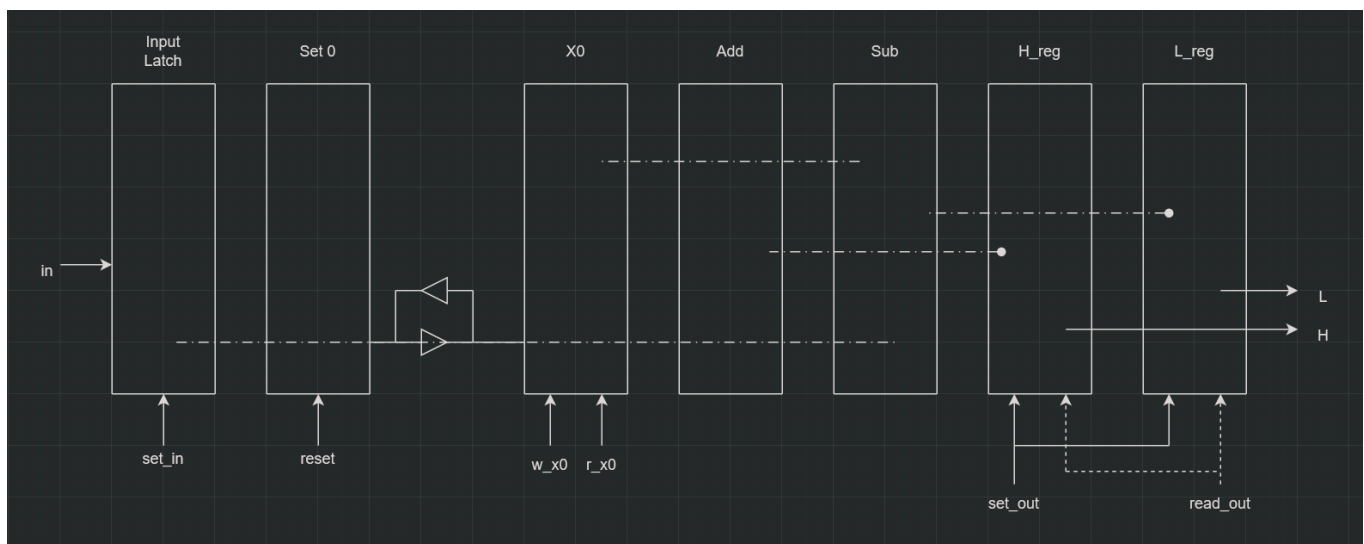
We are implementing the Discrete Wavelet Transformation, specifically working off of the Haar Mother Wavelets. We are doing a 1-Dimensional Transformation, involving a real-time stream of data. We'll call the stream of data **x**. Then for each number within x, the waveform will be filtered by its high frequency component **H**, and its lower frequency component **L** with the following mathematical equation:

$$H = \frac{x_{2i+1} - x_{(2i)}}{2}$$

$$L = \frac{x_{2i+1} + x_{(2i)}}{2}$$

As the data stream goes on, H and L each become their own serial data stream.

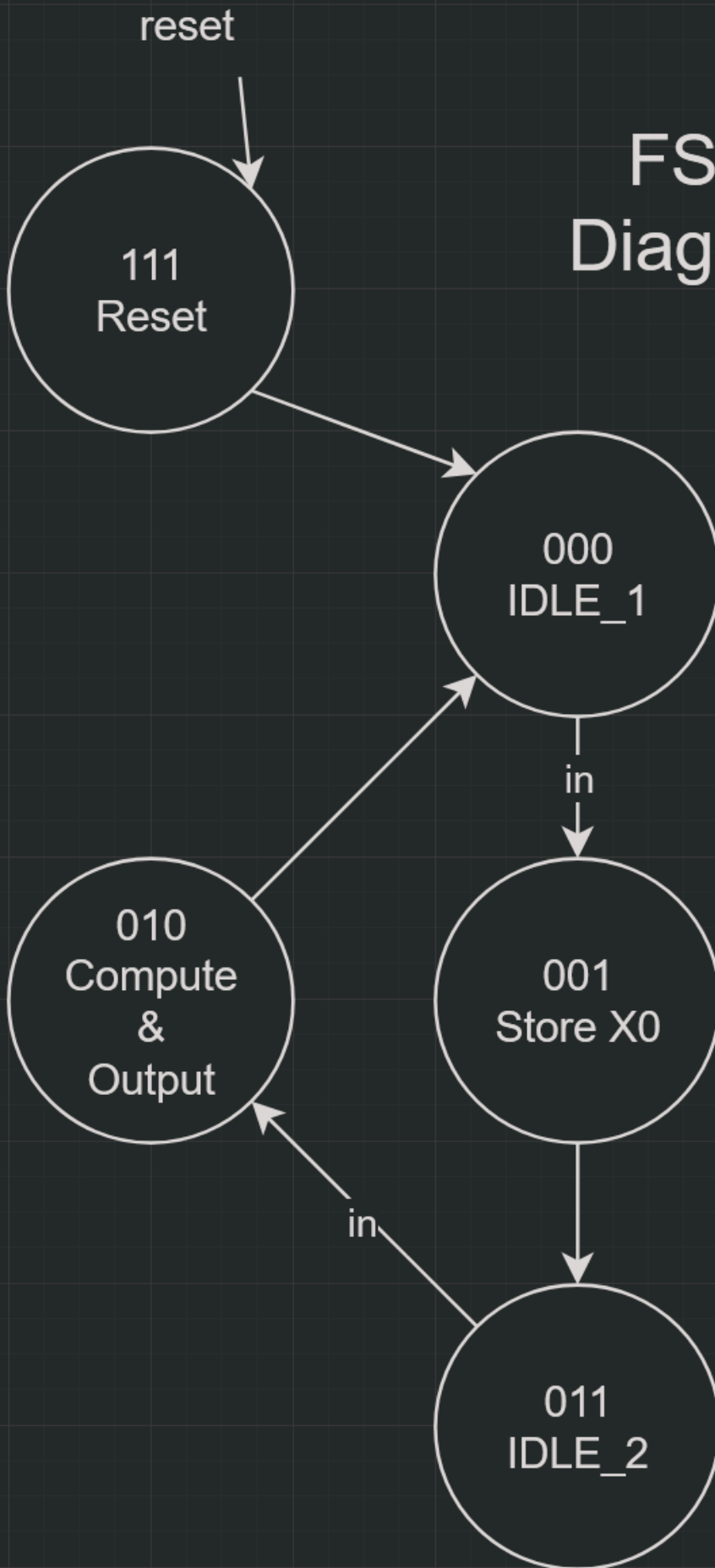
Architecture



Like the math, the datapath is fairly straightforward. One register is reserved to store the x_{i-1} unit, and add and sub blocks are made to do the calculations for both H and L in parallel. Within the path between the Addition/Subtraction and the output registers are wired to provide a right arithmetic shift, which does the equivalent of a division by 2. The two output registers are then outputted to different parts of the chip.

State Machine Diagram

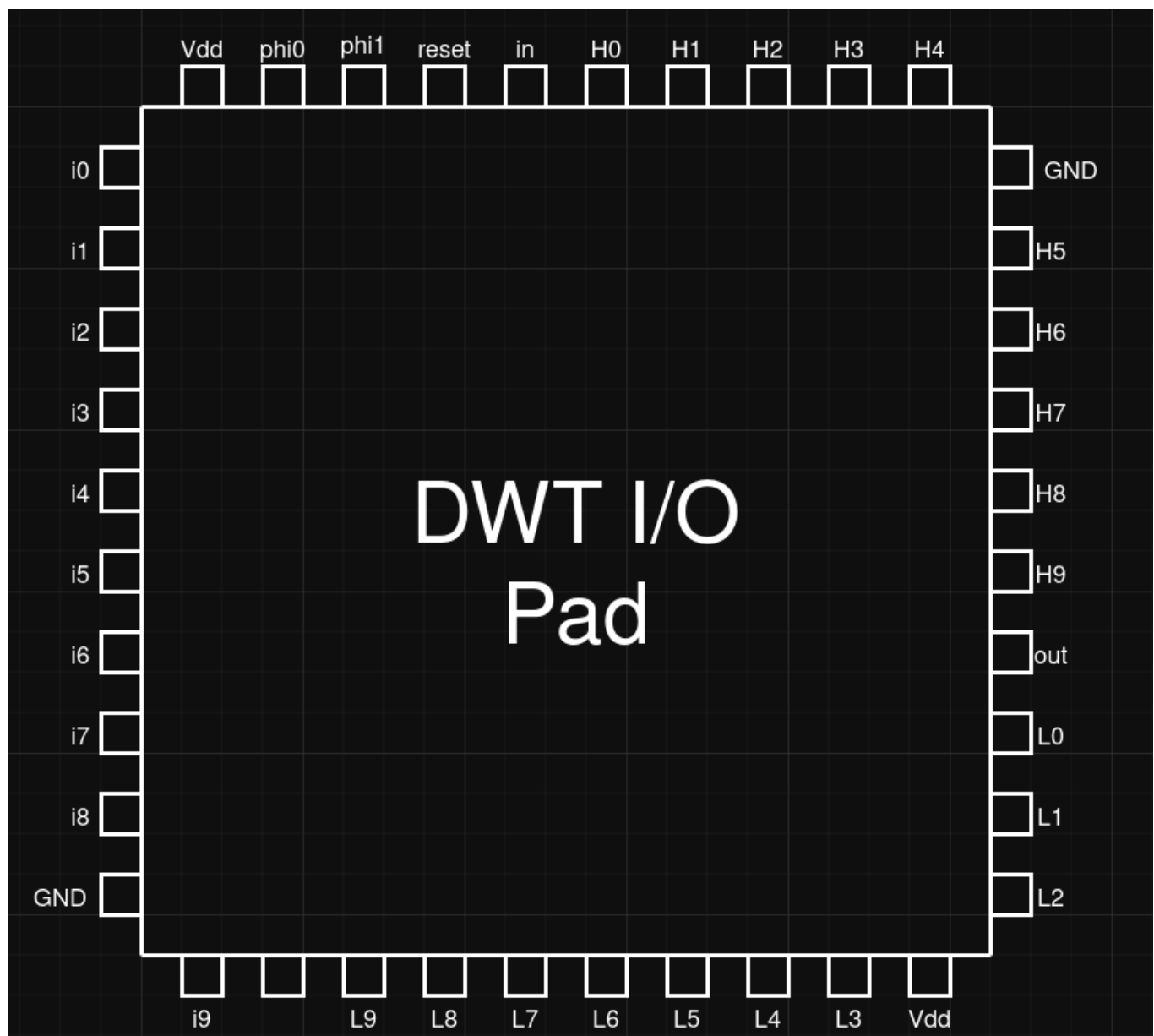
FSM Diagram



The machine first stays within an IDLE state at 000, where it simply waits for any valid input signal. When an input first comes in, it stores that within the x_{i-1} register to be used later. It then enters a second IDLE state, very similar to the first. Then when the next input comes it uses its two stored values to compute and output H and L and an out_valid signal, before returning back to state 000. At any time within this process if reset is brought high, the machine will reset itself.

This process allows the chip to operate pseudo-asynchronously. As long as the external data stream is not faster than two clock cycles, none of the inputs should be lost, and luckily it's fairly easy to keep the clock speed high, as the main bottleneck is one subtraction unit, with every other operation being much quicker. This allows a device that is fairly flexible in use with different sensors.

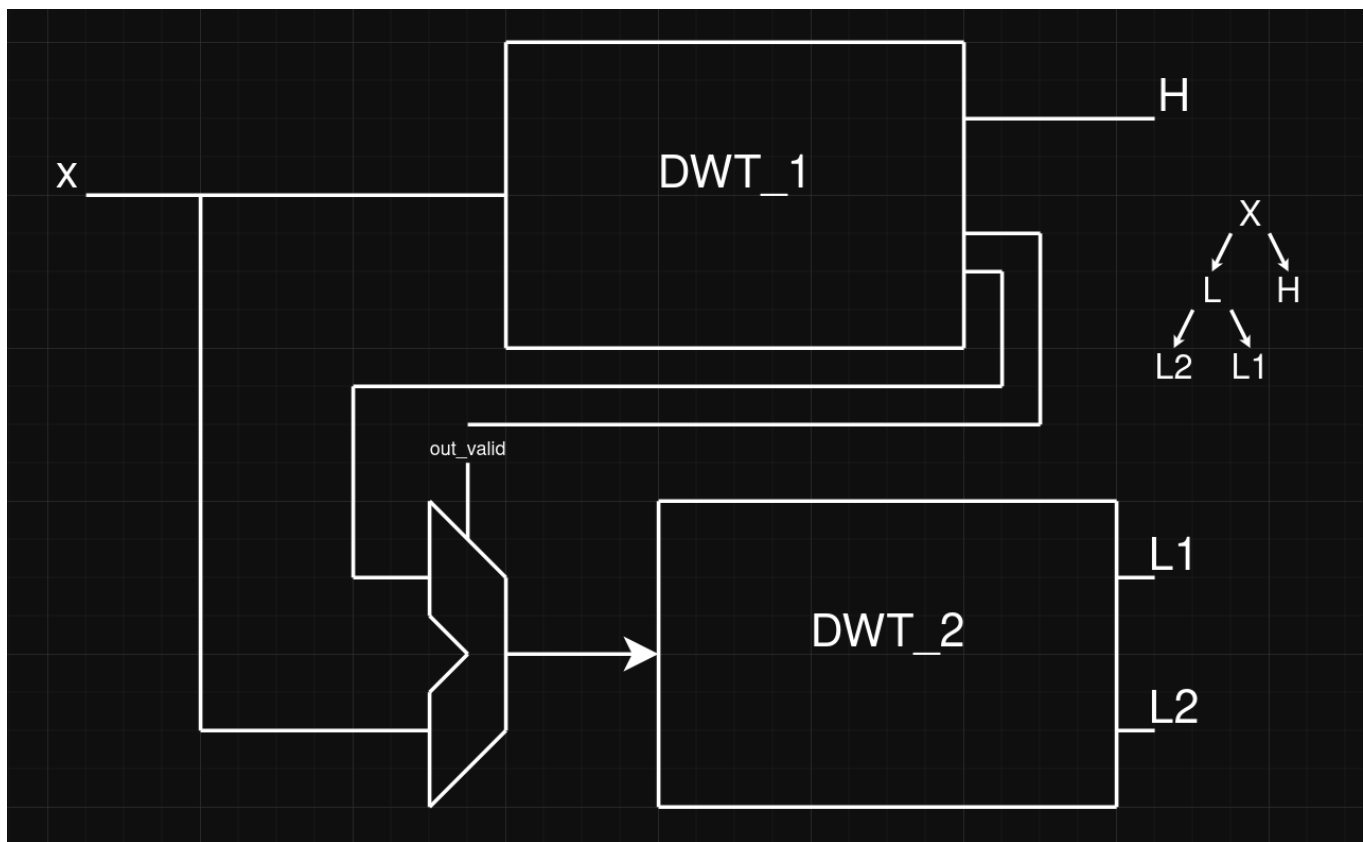
I/O Pad



Above is the diagram of the in and out pins from the Datapad. All inputs and outputs are coded with 10 bits. i_0 to i_9 are the inputs of x , H_0 to H_9 are the Higher Filtered Parts of the Signal x , and L_0 to L_9 are the Low Passed Parts of the Signal x . The signal "in" is used to verify a valid input, it's 1 when a valid input is being sent through, and 0 otherwise. The same is done for the "out" signal, but for outputs instead.

Next Steps

With this processor, only a single-layer transformation is done. Whilst this can be satisfactory for some applications, it can often be better to reiterate the L factor of data multiple times in order to achieve a multi-octave tree. Our architecture can theoretically be iterated upon infinitely, with the main downside of each layer taking more and more time and inputs to calculate, and requiring more power draw from the input buss.



Each input arrives within the chip on ϕ_0 , and each output is made on ϕ_1 . This complementary system allows the signals to guarantee stability between each other before being sampled. Additionally because the output ideally takes double the time of an x input, allowing DWT_2 to alternate between taking a new x value and L as-needed with minimal added control logic, only requiring a MUX that utilizes the out_valid output c-signal.

Final Design and Conclusions



Overall, the final design successfully implements a real-time, single-level 1D Haar Discrete Wavelet Transform and operates as intended based on the defined mathematical model and architectural constraints. Outside of the future steps mentioned above, further improvements could be made towards both the flip-flop registers and summation blocks, further increasing the possible clock speed. While the current implementation performs a single-stage decomposition, the modular nature of the design allows for straightforward extension into a multi-level wavelet tree by iteratively processing the low-frequency output. This scalability, combined with the complementary clocking scheme, suggests that deeper transformations can be achieved with minimal additional control logic.